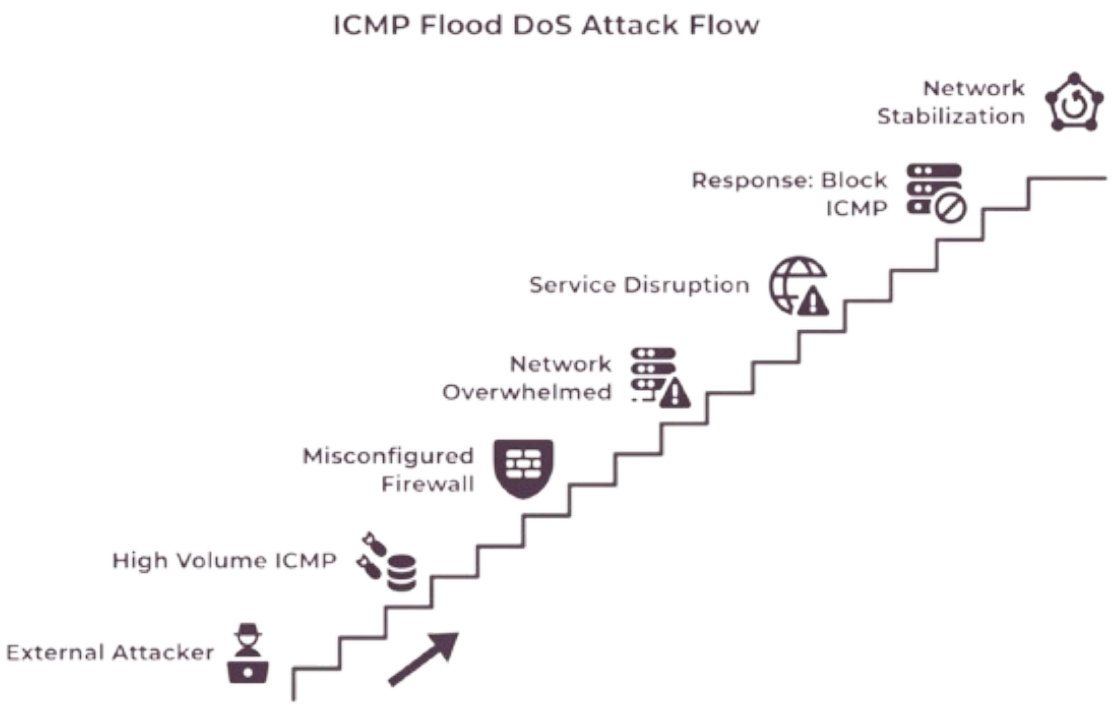
Network Security Incident Report

ICMP Flood DoS Attack

This report presents an educational analysis of an ICMP flood DoS scenario.

1. Summary

A Denial-of-Service (DoS) attack occurred when an external malicious actor targeted the internal network using an ICMP packet flood. The attack involved a high volume of incoming ICMP packets, overwhelming network resources and rendering internal network services unavailable for approximately two hours. Initial investigation indicates that an improperly configured firewall permitted unrestricted inbound ICMP traffic from external networks. During the active incident, response personnel acted to contain the attack by blocking inbound ICMP traffic at the perimeter and temporarily filtering non-critical network services. Following traffic blocking, network throughput stabilized, enabling full restoration of business operations. Recommendations for permanent architectural hardening and long-term preventive controls are detailed in subsequent sections.



2. Identify

- **Attack Type:** Denial-of-Service (DoS) via ICMP Packet Flood / Ping Flood.
- **Affected Systems:** Internal network connectivity and network services required for daily business operations.
- **Root Cause:** An improperly configured firewall that permitted excessive inbound ICMP traffic, allowing the attack to overwhelm network resources.
- **Scope & Impact:** Internal network services were disrupted for approximately two hours. Further log analysis and forensic investigation are recommended to determine whether data confidentiality or integrity was affected.

3. Protect

To mitigate future DoS vulnerabilities and strengthen perimeter resilience, the organization should evaluate and implement the following recommended technical safeguards:

- **Firewall Re-configuration & ICMP Rate Limiting:** Review perimeter firewall policies to restrict unnecessary inbound ICMP traffic, apply appropriate rate limits, and preserve ICMP functions required for legitimate network operations.
- **Source IP Verification & Anti-Spoofing:** Implement reverse path forwarding (RPF) and ingress filtering at boundary routers to block spoofed external IP packets before entering internal subnets.
- **Network Intrusion Prevention System (IPS) Evaluation:** Deploy signature-based and anomaly-based IPS controls to automatically detect, alert, and drop excessive volumetric traffic surges.
- **Network Segmentation:** Segment core network zones (corporate endpoints, internal servers, DMZ) using VLANs and micro-segmentation to minimize the potential blast radius of internal or perimeter disruptions.

4. Detect

To improve threat visibility and reduce mean time to detect (MTTD), the security team should consider deploying and configuring the following monitoring capabilities:

- **Continuous Network Flow Monitoring:** Deploy real-time flow monitoring tools (NetFlow/IPFIX) to define baseline network activity and spot volumetric anomalies promptly.
- **SIEM & Firewall Log Integration:** Centralize firewalls, router, and perimeter event logs into a SIEM platform to perform automated correlation and spike detection.
- **Intrusion Detection System (IDS) Sensors:** Configure IDS sensors at external boundary points to flag ICMP flood signatures and anomalous traffic bursts.

- **Automated SOC Alerting Thresholds:** Establish high-priority alerts to notify security analysts immediately whenever ICMP packet counts exceed baseline operational thresholds.

5. Respond

In the event of future DoS or network volumetric incidents, the organization will execute the following Incident Response Plan (IRP) procedures:

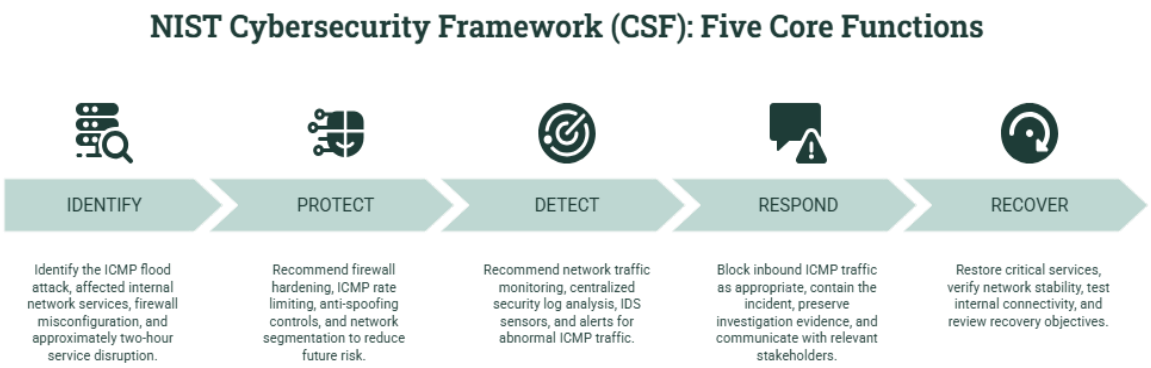
- **Containment & Filtering:** Immediately trigger automated firewall filtering and rate limiting on suspected attack ports and protocols.
- **Network Segment Isolation:** Temporarily isolate affected network segments to prevent traffic degradation from spreading to core business functions.
- **Upstream Mitigation:** Coordinate with the Internet Service Provider (ISP) to evaluate upstream traffic filtering or DoS mitigation services if incoming attack traffic exceeds the organization's available network capacity.
- **Evidence Preservation & Forensics:** Preserve router/firewall logs, PCAP traffic captures, and SIEM alerts for forensic analysis and root-cause verification.
- **Stakeholder Communication:** Notify IT management, SOC leads, and business unit leaders according to established escalation matrices.
- **Post-Incident Review:** Conduct a post-mortem "lessons learned" review to refine response playbooks and update technical controls.

6. Recover

Recovery steps to ensure complete restoration of services and system integrity:

- **Prioritized Service Restoration:** Restore network routing and critical internal services sequentially to manage traffic stability.
- **Network Stability Verification:** Perform health checks on gateway firewalls, routers, and bandwidth usage to confirm traffic levels have returned to normal baseline.
- **Security Validation:** Audit active firewall rules and IPS configurations to verify that mitigations are actively enforcing protection.

- **Functional End-to-End Testing:** Validate internal service connectivity and user access across all business units prior to declaring incident closure.
- **RTO/RPO Metrics Review:** Review the Recovery Time Objective (RTO) for critical network services and validate actual restoration time against the established target. Review Recovery Point Objectives (RPOs) for systems where data or configuration recovery is relevant.



7. Reflections / Notes

Applying the NIST Cybersecurity Framework (CSF) provides a structured, repeatable method for evaluating security incidents across all lifecycle phases (Identify, Protect, Detect, Respond, Recover). This incident highlights the critical importance of continuous network monitoring, strict firewall governance, and regular rule audits to eliminate configuration flaws before they are leveraged by attackers. This case study applies incident analysis, root cause identification, defensive control selection, and recovery planning to a network availability incident.